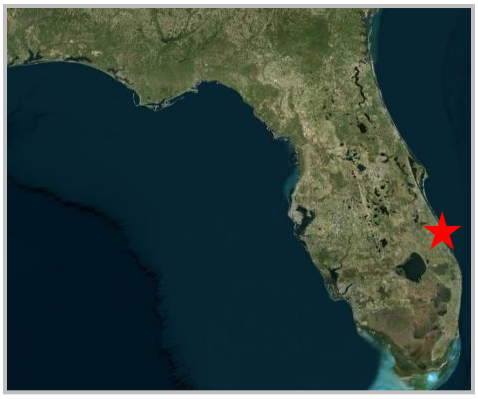
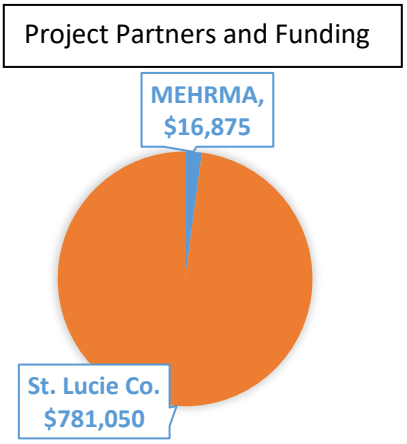
Harbor Branch Mangrove Marsh Restoration Project in St. Lucie County, Florida

FY 17/18





Project Description

In 2010, the Marine and Estuarine Sub-section of FWC, in partnership with St. Lucie County, applied for and was awarded a $1M grant from the US Fish and Wildlife Service (USFWS) through their National Coastal Wetland Conservation Grant Program to implement a 178-acre wetland restoration project in the Indian River Lagoon (IRL) system.

The overarching goal of the restoration project was to conserve and restore tidal connectivity to 178 acres of impaired estuarine intertidal forested habitat (mangroves) that were impounded in the 1950's and 1960's for mosquito control. This was done by restoring hydrologic connectivity to the wetland through the stabilization of an existing but severely eroded dike and the installation of culverts to restore tidal exchange between the wetland and the IRL.



Figure 1: Newly installed riprap along the dike.

FWC contracted St. Lucie County Mosquito Control District (SLC MCD), the project designer, to complete construction of this project. The project was successfully completed in February 2016 by stabilizing the dike and installing 18 culverts.

The FY 17/18 MEHRMA award was originally intended to purchase and place limerock to reinforce priority shoreline areas to adequately protect the project perimeter from wind and wave action. Due to unexpected construction challenges and unanticipated additional expenses, the original project construction fell short of providing adequate stabilization of the culverts and the dike (on the IRL side). The addition of limerock was intended to provide extra security against future breaches of the dike or de-stabilization of culverts which would compromise the effective tidal exchange established by the original project. At the end of FY 16/17, MEHRMA funding became available and was used to purchase 300 cy of limerock that was stockpiled on site, to be placed when MEHRMA 17/18 funding became available.

Prior to implementation of the FY 17/18 MEHRMA project, in September 2017, Hurricane Irma passed over southern Florida and caused extensive damage to the dike. As a result, repairs and maintenance not previously planned were required. FEMA became a project partner and will ultimately reimburse St. Lucie County for materials that were purchased and placed to repair the dike and the road damaged by the storm. Following Irma, much of the stabilization work that was to be implemented by FWC, became part of the FEMA emergency repair project. Thus, MEHRMA funding was used only to place FWC purchased riprap on areas of the dike that were not impacted by Hurricane Irma, but were identified as needing reinforcement prior to Irma. Specifically, the riprap was placed around 18 culverts and on 150 linear feet of shoreline.

Benefits

- Stabilize culverts that were installed to return tidal flow to a 178-acre wetland
- Stabilize dike frontage facing the IRL around the wetland to prevent dike breaches and to protect 178-acre wetland from erosion from high water events caused by storms and king tides
- Protect 178 acres of restored mangrove wetland connected to the Indian River Lagoon
- Increase resilience of the mangrove marsh to sea level rise

FWC Strategic Plan Elements Supported

Theme 1, Goal 2: Ensure that sufficient habitats exist to support healthy and diverse fish and wildlife populations.

Strategies:

- Protect lands and waters critical to sustaining healthy and diverse fish and wildlife populations through diverse incentive programs.
- Manage habitats to sustain healthy and diverse fish and wildlife populations.

Acreage restored by habitat type (footprint)

This project reinforced the entire 2-mile dike that runs the length of the IRL shoreline. FWC's funding was specifically used to place 300 cu yds of limerock (purchased by FWC with MEHRMA funds in FY 16/17) on 18 culverts and 150 linear feet of unarmored shoreline.

Acreage affected by restoration activities: 178 acres